ECN 711: Government in OG Economy

Gustavo Ventura

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- Let $G(t)$ be goods consumed by the government at t.
- Government's budget constraint:
$$G(t) = \sum_{h=1}^{N(t)} \tau_t^h(t) + \sum_{h=1}^{N(t-1)} \tau_{t-1}^h(t).$$

- Given taxation, lifetime budget constraint is now:

$$c_t^h(t) + \frac{c_t^h(t+1)}{R(t)} \leq w_t^h(t) - \tau_t^h(t) + \frac{w_t^h(t+1) - \tau_t^h(t+1)}{R(t)}.$$

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- Individual decisions are characterized by still characterized by the same first-order conditions. Why?
- What is an equilibrium in these circumstances?

Let the tax sequence T be

$$T \equiv \left\{ \left(\tau_{t-1}^{h}(t) \right)_{h=1}^{N(t-1)}, \left(\tau_{t}^{h}(t) \right)_{h=1}^{N(t)} \right\}_{t=1}^{\infty}$$

We can now define a competitive equilibrium.

Equilibrium Definition

A competitive equilibrium is a tax sequence $\widetilde{T}$, a sequence of government consumption $\left\{\widetilde{G}(t)\right\}_{t=1}^{\infty}$, a consumption allocation $\widetilde{C}$, a sequence of lending/borrowing decisions, $\left\{\left(\widetilde{l}^h(t)\right)_{h=1}^{N(t)}\right\}_{t=1}^{\infty}$, and interest rates $\left\{\widetilde{R}(t)\right\}_{t=1}^{\infty}$ such that

(1) Given $\widetilde{T}$ and $\left\{\widetilde{R}(t)\right\}_{t=1}^{\infty}$, individuals maximize utility and choose the relevant elements of $\widetilde{C}$.

(2) Market clearing, i.e. for all $t \geq 1$

$$\sum_{h=1}^{N(t)} \widetilde{l}^h(t) = 0.$$

(3) The government budget balances for all $t \geq 1$

$$\sum_{h=1}^{N(t)} \widetilde{\tau}_t^h(t) + \sum_{h=1}^{N(t-1)} \widetilde{\tau}_{t-1}^h(t) = \widetilde{G}(t).$$

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- Exercise: show that an allocation $\widetilde{C}$ and a path of government consumption are feasible *if and only if* the market for consumption loans clears in all dates.

Exercise: intergenerational transfers

- Identical individuals within a cohort of size N. Preferences:

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- How are equilibrium interest rates and allocations affected?

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- Equilibrium interest rate is higher than in the absence of taxes/transfers. Why? What are the implications of population growth?

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- Endowments: ω_y and ω_o.
- Only young pay taxes, $\tau_t(t)$. Government consumption fluctuates:

  $G(t) = G_H$ if t is even. $G(t) = G_L$ if t is odd. $G_H > G_L$.

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- Why are equilibrium interest rates higher when government consumption is high?

Public Debt

- We now assume that the government can issue debt in each date, $B(t)$, at the price $p(t)$. The budget constraint of the government is now:

$$\sum_{h=1}^{N(t)} \tau_t^h(t) + \sum_{h=1}^{N(t-1)} \tau_{t-1}^h(t) + p(t)B(t) = G(t) + B(t-1)$$

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- Individual budget constraints are now:

$$c_t^h(t) = w_t^h(t) - \tau_t^h(t) - l^h(t) - p(t)b^h(t), \tag{1}$$

$$c_t^h(t+1) = w_t^h(t+1) - \tau_t^h(t+1) + R(t)l^h(t) + b^h(t), \tag{2}$$

Public Debt

- (1) and (2) imply lifetime constraint

$$
\begin{aligned}
c_t^h(t) + \frac{c_t^h(t+1)}{R(t)} &= w_t^h(t) - \tau_t^h(t) + \frac{w_t^h(t+1) - \tau_t^h(t+1)}{R(t)} \\
&- b^h(t)\left[p(t) - \frac{1}{R(t)}\right].
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- Note that in equilibrium, it has to be that $p(t) = 1/R(t)$. Why?

  Lifetime budget constraint is the same as before (last term disappears!).

Equilibrium with Public Debt

- Hence, individuals are indifferent between holding debt or consumption loans. Let

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denote what they are willing to lend (or borrow) to the government or others.

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- Remark: Note that there is no **net** borrowing and lending in any equilibrium. Why?

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- $G(t) = 0$ all t. Unexpectedly, the government transfers $\triangle = 1/2$ to current old at t_0, financed with debt, to be fully repaid by the young at $t_0 + 1$.
- Obtain the path of interest rates in this economy. Show that $R(t) = 1/2$ if $t < t_0$ or $t > t_0 + 1$. Further, $R(t_0) = 1$ and $R(t_0 + 1) = 2/3$.